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United States Patent	12404702
Kind Code	B1
Date of Patent	September 02, 2025
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Anti-theft device having an electromechanically actuated arm

Abstract

An anti-theft device for securing an article of merchandise against unauthorized removal from a display counter. The anti-theft device includes a plurality of arms configured to immobilize the article of merchandise relative to a tray. At least one arm is operated by an electromechanical release mechanism between closed and opened positions. When the electromechanically actuated arm is in the closed position, the article of merchandise is immobilized relative to the tray. When the electromechanically actuated arm is transitioned into the open position, the article of merchandise can be removed from the tray. The anti-theft device may include a locking mechanism configured to transition between a locked configuration, in which the arms are immobilized relative to the tray, and an unlocked configuration, in which the arms can be extended or retracted relative to the tray.

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Appl. No.: **18/675988**

Filed: **May 28, 2024**

Related U.S. Application Data

continuation parent-doc US 17851913 20220628 US 11993959 20240528 child-doc US 18675988
continuation parent-doc US 17545542 20211208 US 11401738 20220802 child-doc US 17851913
us-provisional-application US 63153506 20210225

Publication Classification

Int. Cl.: **E05B73/00** (20060101); **E05B41/00** (20060101); **E05B47/00** (20060101); **G07C9/00** (20200101); **G08B13/14** (20060101)

U.S. Cl.:

CPC **E05B73/0082** (20130101); **E05B41/00** (20130101); **E05B47/0012** (20130101); **G07C9/00182** (20130101); **G08B13/1472** (20130101); E05B2047/0016 (20130101); E05B2047/0017 (20130101)

Field of Classification Search

CPC: E05B (73/00); E05B (41/00); E05B (47/0012); E05B (2047/0016); E05B (2047/0017); G07C (9/00182); G07B (13/1472)

USPC: 70/58-62

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Background/Summary

PRIORITY CLAIM (1) This non-provisional patent application is a continuation of and claims priority to the U.S. Non-Provisional patent application Ser. No. 17/851,913 filed on Jun. 28, 2022, which is a continuation of and claims priority to the U.S. Non-Provisional patent application Ser. No. 17/545,542 filed on Dec. 8, 2021, which is a continuation of and claims priority to U.S. Provisional Patent Application No. 63/153,506 filed on Feb. 25, 2021.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) This invention relates to merchandise anti-theft devices. More specifically, it relates to an anti-theft device having an electromechanical release mechanism that selectively locks or releases an article of merchandise.

2. Brief Description of the Related Art

(2) Retailers often prefer to present their merchandise to customers in a way that allows the customers to touch, inspect, and otherwise interact with the products at a display counter. Many merchandise items, especially portable electronic devices, are relatively expensive and, therefore, are under a serious threat of theft. Retailers often face a dilemma pertaining to how to interactively display their merchandise to attract customers and increase sales, while, at the same time, safeguarding the merchandise against theft.

(3) Currently available anti-theft solutions generally involve obtrusive, bulky, heavy, and aesthetically unattractive devices such as brackets, steel cables, locks, and casings. Although these security measures may effectively protect against theft, they have a negative effect on the customer shopping experience by discouraging interaction with products and may ruin the overall ambiance of a retail store.

(4) When a customer decides to go to a retail store to have a firsthand experience with an electronic gadget, that customer generally desires to preview the experience of owning the electronic gadget. However, most state-of-the-art anti-theft devices are not designed to provide the customer with a pure, unadulterated interaction with the electronic gadget. Unfortunately, because most anti-theft devices stay attached to the electronic gadget during the entirety of a customer interaction, these traditional anti-theft devices can diminish the quality of customer experience in multiple ways. First, when an anti-theft device is attached to an electronic gadget, the customer cannot feel the true weight of the electronic gadget. Second, traditional anti-theft devices make electronic gadgets appear much bulkier and less aesthetically pleasing. Third, anti-theft devices can ruin the ergonomics of the design. These flaws undermine engineering and marketing efforts that manufacturers of electronic gadgets undertake to make their devices slimmer, lighter, and more pleasing to handle.

(5) Consequently, while it is crucial for retailers to protect their expensive merchandise against theft, the retailers may wish to enable trusted customers to interact with their merchandise in its pure form, unaltered by an attached anti-theft device. In such circumstances, retailers and their trusted customers would significantly benefit from an anti-theft device that can quickly release the secured merchandise. Furthermore, to avoid ruining the high-tech aesthetics of an electronic gadget by unlocking a lock to remove a bracket, it would be beneficial for the anti-theft device to operate via an electromechanical mechanism that seamlessly releases the secured electronic gadget in a visually appealing manner, commensurate with the high value of the electronic gadget being displayed.

(6) Accordingly, there exists an unresolved need for an effective anti-theft device having an electromechanical release mechanism.

SUMMARY OF THE INVENTION

(7) The unresolved need stated above is now met by a novel and non-obvious invention disclosed and claimed herein. In an embodiment, the invention pertains to an anti-theft device for securing an article of merchandise—for example, an electronic gadget such as a smartphone or a tablet. The anti-theft device has a pedestal configured to be affixed to a support surface. The pedestal has a tray, wherein the tray is configured to support the article of merchandise thereon. As used herein, the term “tray” refers to a surface on which the article of merchandise is positioned while secured by the anti-theft device. The tray may be a surface of the pedestal itself, or the tray can be a separate component configured to couple to the pedestal.

(8) The anti-theft device has a plurality of arms configured to immobilize the article of merchandise relative to the tray. At least one of the arms is configured to be movable relative to the tray using an electromechanical actuation mechanism. This electromechanically actuated arm has a closed position in which the article of merchandise is immobilized relative to the tray. The electromechanically actuated arm also has an open position in which the article of merchandise can be removed from the tray.

(9) The anti-theft device may further include a locking block movingly disposed within the pedestal. The locking block has a first position relative to the pedestal in which the locking block engages the electromechanically actuated arm to transition the electromechanically actuated arm into the closed position and immobilize it in that closed position. The locking block also has a second position relative to the pedestal in which the locking block releases the electromechanically actuated arm, allowing it to transition into the open position.
(10) A motor is positioned within the pedestal and is operably connected to the locking block. The motor may be an electric motor, a pneumatic motor, or any other type of motor suitable for rotating a driving screw. For ease of reference, the exemplary embodiment described herein refers to an electric motor, but this description should not be interpreted as exclusionary of other motor types.
(11) Actuation of the electric motor selectively moves the locking block between the first position and the second position. In this manner, selective actuation of the electric motor closes the electromechanically actuated arm to immobilize the article of merchandise relative to the tray or opens the electromechanically actuated arm to release the article of merchandise from the tray.
(12) In an embodiment, the electric motor is operably connected to the locking block via a worm drive. The worm drive prevents the electromechanically actuated arm from being transitioned from the closed position into the open position by application of a force or a moment onto the electromechanically actuated arm.
(13) The electromechanically actuated arm may have a stem pivotally disposed within the pedestal, wherein the stem is configured to pivot about a pivot axis, transitioning the arm between the closed position and the open position. The locking block may be configured to exert a force onto the stem of the arm wherein a point of contact between the locking block and the stem of the arm is offset relative to the pivot axis, thereby causing the stem of the first arm to rotate about the pivot axis, transitioning the electromechanically actuated arm into the closed position. The electromechanically actuated arm may be biased toward the open position, such that retracting of the locking block from the stem of the electromechanically actuated arm causes the arm to automatically transition into the open position.
(14) In an embodiment of the anti-theft device, when the electromechanically actuated arm is in the open position, the article of merchandise can be removed from the tray solely in a direction parallel to the tray. At least some of the arms are configured to be extended and retracted in a plane parallel to the tray. Specifically, the stem of the electromechanically actuated arm may be configured to extend and retract relative to the tray. The adjustable arms may have teeth disposed thereon, and the anti-theft device may further include a locking member configured to selectively engage the teeth to immobilize the arms relative to the tray or to selectively disengage the teeth thereby enabling the arms to be adjusted relative to the tray. The locking member may be accessible via a port disposed on the pedestal. When the locking block is in the first position (in which the electromechanically actuated arm is opened), the access to the locking member via the port is unobstructed. However, when the locking member is in the second position (in which the electromechanically actuated arm is closed), the locking member is inaccessible via the port, whereby the locking member cannot be operated to disengage the teeth of the arms.
(15) In an embodiment, an inductive coil is disposed underneath the tray. The inductive coil may be configured to wirelessly supply power to the article of merchandise. In this manner, the inductive coil positioned underneath the tray may be configured to detect presence of the inductive coil housed within the article of merchandise. When the inductive coil underneath the tray detects the inductive coil of the article of merchandise, the motor is automatically actuated to transition the first arm into the closed position. In an embodiment, the anti-theft device may be configured to identify a type or a model of the article of merchandise positioned on the tray. In this embodiment, the electromechanical mechanism is actuated only when the correct type/model of the article of merchandise is placed onto the tray.
(16) Another feature of the anti-theft device is that the electric motor may be configured to cease

operation responsive to detecting a force being applied to the electromechanically actuated arm while it is transitioning from the open position into the closed position. Furthermore, the anti-theft device may be configured to generate an alarm responsive to detecting a force being applied to the electromechanically actuated arm while the electromechanically actuated arm is transitioning from the open position into the closed position.
(17) The electric motor of the electromechanical actuation mechanism may be configured to be actuated to transition the first arm into the open position responsive to receiving a wireless signal from a designated key fob for or a remote control. In an embodiment, while the electromechanically actuated arm remains in the open position, the designated remote control cannot be used to actuate another electric motor of another anti-theft device.
(18) A photo-interrupter, proximity sensor, limit switch, or other position detection device may be used to determine whether the locking block is in the first position or in the second position. The anti-theft device may be configured to generate an alarm if the article of merchandise is removed from the tray and is not returned to the tray prior to expiration of a designated time period.
(19) The anti-theft device may further include a light emitting device disposed within the pedestal. The light emitting device may be configured to project a light via a lens onto a designated surface. A color of the emitted light may be used to convey information pertaining to the anti-theft device.

Description

DESCRIPTION OF THE DRAWINGS
(1) For a fuller understanding of the invention, reference should be made to the following detailed description, taken in connection with the accompanying drawings, in which:
(2) FIG. **1**A is a perspective view of the anti-theft device in an unlocked configuration.
(3) FIG. **1**B is a perspective view of the anti-theft device in a locked configuration.
(4) FIG. **2** is a rear-right view of an embodiment of the anti-theft device in a locked configuration.
(5) FIG. **3**A is a top cross-sectional view depicting the arm-locking mechanism in an unlocked state.
(6) FIG. **3**B is a detail view depicting the arm-locking mechanism in an unlocked state.
(7) FIG. **4**A is a top cross-sectional view depicting the arm-locking mechanism in a locked state.
(8) FIG. **4**B is a detail view depicting the arm-locking mechanism in a locked state.
(9) FIG. **5**A is a perspective view of the electromechanical release mechanism with the electromechanically actuated arm in the open position.
(10) FIG. **5**B is a perspective view depicting the mechanically adjustable arms and the electromechanical release mechanism with the electromechanically actuated arm in the open position.
(11) FIG. **5**C is a detail view of the electromechanical release mechanism with the electromechanically actuated arm in the open position.
(12) FIG. **6**A is a perspective view of the electromechanical release mechanism with the electromechanically actuated arm in the closed position.
(13) FIG. **6**B is a perspective view depicting the mechanically adjustable arms and the electromechanical release mechanism with the electromechanically actuated arm in the closed position.
(14) FIG. **6**C is a detail view of the electromechanical release mechanism with the electromechanically actuated arm in the closed position.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT
(15) In the following detailed description of the preferred embodiment, reference is made to the accompanying drawings, which form a part hereof, and within which specific embodiments are shown by way of illustration by which the invention may be practiced. It is to be understood that

other embodiments may be utilized and structural changes may be made without departing from the scope of the invention.

(16) FIGS. **1**A and **1**B depict an anti-theft device **10** according to an embodiment of the invention. In this embodiment, anti-theft device **10** has a tray **12**. As used herein, the term “tray” refers to a surface on which the article of merchandise **14** is positioned while secured by anti-theft device **10**. In FIG. **1**A, tray **12** is depicted as a surface of a separate component coupled to a pedestal **20**. Alternatively, tray **12** may be a surface of pedestal **20** itself, on which article of merchandise **14** is configured to rest. The term “tray” covers both of these implementations.

(17) Next, FIGS. **1**A and **1**B depict a plurality of arms configured to secure an article of merchandise to tray **12**. Each arm can either be fixed, manually adjustable, or electromechanically actuated. The embodiment depicted in FIGS. **1**A and **1**B has three manually adjustable arms **16** and one electromechanically actuated arm **18**. Manually adjustable arms **16** and electromechanically actuated arm **18** are configured to extend and retract relative to tray **12**, as explained in more detail below. The number and arrangement manually adjustable arms **16** and electromechanically adjustable arms **18** can vary across various embodiments of the invention, depending on the type and dimensions of article of merchandise **14** being secured. Furthermore, some embodiments of the invention may include one or more fixed arms that do not extend and retract relative to tray **12**.

(18) FIG. **1**A depicts anti-theft device **10** in its unlocked configuration. In this configuration, electromechanically actuated arm **18** is rotated away from tray **12**, enabling removal of article of merchandise **14**. FIG. **1**A depicts electromechanically actuated arm **18** in an open position, in which electromechanically actuated arm **18** is rotated away from the plane of tray **12**. In this manner, article of merchandise **14** can be removed from tray **12** by laterally sliding article of merchandise **14** out of the grips of arms **16**.

(19) FIG. **1**B depicts a locked configuration of anti-theft device **10** in which article of merchandise **14** is immobilized relative to tray **12** by manually adjustable arms **16** and electromechanically actuated arm **18**. In the embodiment depicted in FIGS. **1**A and **1**B, manually adjustable arms **16** have grips configured to receive edges of article of merchandise **14**. The grips partially extend over the front surface of article of merchandise **14**, thereby immobilizing article of merchandise **14** against being lifted upward (in the direction of z-axis) from tray **12**. Furthermore, FIG. **1**B depicts that manually adjustable arms **16** and electromechanically actuated arm **18** engage all four edges of article of merchandise **14**, thereby immobilizing article of merchandise **14** against lateral movement in the x-y plane relative to tray **12**.

(20) FIGS. **1**A and **1**B further depict that tray **12** is coupled to or integrated into pedestal **20**. Pedestal **20** is configured to be securely mounted onto a display counter in a retail store. Thus, when anti-theft device **10** is in the locked configuration, article of merchandise **14** cannot be removed from the display counter.

(21) In an embodiment, a wireless charging device may be disposed underneath tray **12** and configured to supply power to article of merchandise **14**. The wireless charging device can include one or more inductive coils configured to interact with a corresponding inductive coil housed within article of merchandise **14**. One or more inductive coils can be arranged underneath tray **12** in a manner that enables wireless power transfer to articles of merchandise **14** of various geometries and having various inductive coil placements. In an embodiment, location of the inductive charging coil underneath tray **12** can be configured to be adjustable. In this manner, the inductive charging coil can be moved (either manually or automatically) into a location relative to tray **12** that corresponds to the location of an inductive charging coil within article of merchandise **14**.

(22) One aspect of anti-theft device **10** is automated transitioning from an unlocked configuration depicted in FIG. **1**A into the locked configuration depicted in FIG. **1**B. One method of accomplishing this is by utilizing the inductive coil positioned underneath tray **12** to detect that article of merchandise **14** has been placed onto tray **12**. When the inductive coil underneath tray **12**

detects presence of the inductive coil of article of merchandise **14**, a processor (which may be housed underneath tray **12**, within pedestal **20**, or a remote location) actuates electric motor **36** (depicted in FIGS. **5**A-C and **6**A-C). (Other types of motors—for example, a pneumatic motor—can be used and fall within the scope of the invention). Electric motor **36** causes electromechanically actuated arm **18** to transition from the open position into the closed position. In this manner, when article of merchandise **14** is placed onto tray **12**, electromechanically actuated arm **18** automatically closes to immobilize article of merchandise **14** relative to tray **12** as depicted in FIG. **1**B. (The structure and functionality of the electromechanically actuation mechanism that closes and opens arm **18** is described in more detail below.)
(23) As described above, FIG. **1**B depicts that the grips of arms **16** prevent article of merchandise **14** from being lifted off tray **12** (in direction of the z-axis) and, because manually adjustable arms **16** and electromechanically actuated arm **18** engage all four edges of article of merchandise **14**, article of merchandise **14** cannot be removed in a lateral direction (in direction of the x-axis and y-axis). Thus, when arm electromechanically actuated **18** is closed, article of merchandise **14** is secured within anti-theft device **10** and cannot be removed therefrom until electromechanically actuated arm **18** is transitioned into an opened position depicted in FIG. **1**A.
(24) In various embodiments of the invention, electromechanically actuated arm **18** may be actuated in various ways. For example, in some embodiments, the electromechanical mechanism responsible for transitioning electromechanically actuated arm **18** into an open position may be configured to be actuated via wireless means, such as near field communication (NFC), radio frequency identification (RFID), an optical signal (for example, infrared (IR), or the like. In this embodiment, a verified customer can be presented with a designated key fob or a remote control that is configured to transmit the predefined wireless signal to open electromechanically actuated arm **18**. Once arm **18** is in the opened position (depicted in FIG. **1**A), the customer can remove article of merchandise **14** from tray **12** by sliding article of merchandise **14** to the right in the x-direction. In an embodiment, electromechanically actuated arm **18** can be configured to be actuated via a wired connection to a control mechanism, which may be triggered by an authenticating system that releases the secured article, as described in more detail below.
(25) In alternative embodiments, the electromechanical release mechanism may be actuated in response to accepting a customer's credit card, an identification card, a driver's license, or a proprietary token, card, or chip. Furthermore, the anti-theft device may be equipped with a mechanism configured to read customer's biometric information (for example, by scanning fingerprints or retina) or to require the customer to input his or her credentials. Upon registering/verifying the customer's identity, receiving a monetary security deposit, and/or verifying that the customer presents a low threat risk, the electromechanical release mechanism may be configured to automatically open electromechanically actuated arm **18**, thereby releasing article of merchandise **14** and granting the trusted customer full access to article of merchandise **14**.
(26) Upon completion of the customer interaction, article of merchandise **14** can be placed back onto tray **12** by laterally sliding article of merchandise **14** into the grips of arms **16**. As disclosed above, when tray **12** detects presence of article of merchandise **14**—for example, by utilizing indicative charging coil—the processor of anti-theft device **10** automatically actuates electric motor **36** that closes electromechanically actuated arm **18**, thereby locking article of merchandise **14** relative to tray **12**.
(27) In an embodiment, anti-theft device **10** may have a timer configured to allocate a predefined time duration for the customer interaction with article of merchandise **14**. When electromechanically actuated arm **18** is opened, a timer starts and does not stop until electromechanically actuated arm **18** is closed again in response to article of merchandise **14** being placed back onto tray **12**. In this manner, if article of merchandise **14** is not replaced back onto tray **12** prior to the expiration of the predefined time duration allocated for the customer interaction, an alarm may be triggered. In this embodiment, because the processor is configured to close

mechanically actuated arm **18** only in response to the inductive coil of tray **12** communicating with the inductive coil of article of merchandise **14**, placing a “dummy” object without an inductive coil onto tray **12** would not bypass the timer-based alarm.
(28) Furthermore, the inductive coil (or another electronic component) underneath tray **12** may be configured to identify the model or a unique, singulated identifier of article of merchandise **14**. In this embodiment, the processor will not actuate the electromechanical mechanism to close electromechanically actuated arm **18** unless the exact model of article of merchandise **14** is placed onto tray **12**. In this manner, the timer-based alarm system cannot be bypassed by placing a different (perhaps less expensive) article of merchandise onto tray **12**, even if that article of merchandise includes an inductive charging coil.
(29) FIGS. **1**A-B depict a light emitting diode (LED) **21** integrated into pedestal **20**. LED **21** may be configured to project a light onto the display surface. The light may have different colors and/or flashing patterns to discretely communicate the status of anti-theft device **10** to the store personnel. For example, the color of the light may change upon authorization of a verified customer, upon expiration of the timer, or in response to any other event. In an embodiment, the light may be configured to illuminate through a lens and provide and enable store personnel to adjust the location where the light is directed or projected.
(30) FIG. **2** depicts anti-theft device **10** in the locked configuration from a perspective of right-back view. FIG. **2** depicts an access port **22** disposed on pedestal **20**. Access port **22** provides access to a locking member **24**, which is configured to selectively immobilize manually adjustable arms **16** and electromechanically actuated arm **18** against lateral adjustment relative to tray **12**. As will be explained in more detail below, when electromechanically actuated arm **18** is in the open position, an authorized user can insert a tool into access port **22** to retract locking member **24** away from adjustable arms **16** and electromechanically actuated arm **18**, so that arms **16** and **18** can be extended or retracted relative to tray **12**. However, when electromechanically actuated arm **18** is in the closed position, the access to locking member **24** via access port **22** is blocked.
(31) FIGS. **3**A-B and **4**A-B depict a locking mechanism disposed underneath tray **12** configured to selectively lock or unlock manually adjustable arms **16** and electromechanically actuated arm **18** so that arms **16** and **18** can be extended or retracted relative tray **12** in the x-y plane. One purpose of this feature is to enable retail store personnel to adjust the distances between arms **16** and **18** to accommodate dimensions of article of merchandise **14** being secured.
(32) FIGS. **3**A-B and **4**A-B depict that the arm-locking mechanism includes locking member **24**, which has one or more teeth configured to interlock with the teeth disposed along the stems of arms **16** and **18**. FIGS. **3**A-B depict locking member **24** retracted away from arms **16** and **18**, such that the teeth of locking member **24** disengage the teeth of arms **16** and **18**. While locking member **24** remains in this retracted position, an authorized user can extend and retract arms **16** and **18** relative to tray **12**. To extend or retract arms **16** and **18** relative to tray **12**, the authorized user simply applies a force to arms **16**/**18** to slide arms **16**/**18** in or out relative to tray **12**. In this manner, anti-theft device **10** can be readily adapted to accommodate a wide variety of articles of merchandise **14** having different dimensions.
(33) FIGS. **4**A-B depict locking member **24** in a locked position, in which the teeth of locking member **24** engage the teeth of arms **16** and **18**. While locking member **24** remains in this locked position, arms **16** and **18** are immobilized against movement in the x-y plane relative to tray **12**. Arms **16** and **18** cannot be extended relative to tray **12** until locking member **24** is retracted away from the teeth of arms **16**/**18**, which is the unlocked position depicted in FIGS. **3**A-B. Locking member **24** may be configured to transition between locked and retracted positions in several ways, including using a screw-threaded actuator, an electromagnet (such as a solenoid), an electrical motor, a mechanical push button, etc. In an embodiment, the locking component may be biased toward the locked configuration.
(34) As an added security measure, to retract locking member **24** away from arms **16** and **18**, the

user must access locking member **24** via access port **22**. When electromechanically actuated arm **18** is in the closed position, access to locking member **24** is blocked, preventing access to locking member **24**. In this manner, arms **16** and **18** cannot be extended or retracted relative to tray **12** unless electromechanically actuated arm **18** is in the open position. Thus, when anti-theft device **10** is in the locked configuration, arms **16** and **18** are immobilized relative to tray **12** and cannot be unlocked without operating the electromechanical mechanism that opens electromechanically actuated arm **18**, as described in more detail below.

(35) Next, FIGS. **5**A-C and **6**A-C depict an exemplary embodiment of the electromechanical mechanism that opens and closes electromechanically actuated arm **18**. FIGS. **5**A-C depict electromechanically actuated arm **18** in an open position, in which the electromechanically actuated arm **18** is rotated away from the plane in which article of merchandise **14** is positioned on tray **12**. FIGS. **6**A-C depict electromechanically actuated arm **18** in a closed position, in which the electromechanically actuated arm **18** intersects the plane in which article of merchandise **14** resides on tray **12** and, therefore, when closed, electromechanically actuated arm **18** impedes removal of article of merchandise **14** from tray **12**.

(36) FIGS. **5**A and **6**A depict that electromechanically actuated arm **18** has a stem **30**, which is configured to rotate about a pivot axis **32**. Next, FIGS. **5**B-C and **6**B-C depict a locking block **28** configured to translate vertically within pedestal **20**. Specifically, FIGS. **5**B-C depict that, when locking block **28** is lowered, electromechanically actuated arm **18** is in the open position, while FIGS. **6**B-C depicts that, when locking block **28** is raised, electromechanically actuated arm **18** is in the closed position.

(37) FIGS. **5**B and **5**C depict that, when locking block **28** is lowered, locking block **28** disengages stem **30** of arm **18**. FIGS. **5**B and **5**C also show that a biasing element **34** exerts a force onto stem **30** of arm **18**, wherein the force is offset from pivot axis **32**. Thus, the biasing force creates a moment causing stem **30** to rotate about pivot axis **32**, transitioning arm **18** into the open position depicted in FIGS. **5**B and **5**C. As discussed above, in this open position, arm **18** does not intersect the plane in which article of merchandise **14** is positioned on tray **12** and, therefore, arm **18** no longer restraints article of merchandise **14** against lateral movement in the x-axis direction relative to tray **12**. Thus, an authorized user (a retail employee or a verified customer) can slide article of merchandise **14** laterally out of the grips of arms **16** to remove article of merchandise **14** from anti-theft device **10** for a customer interaction.

(38) As explained above, when article of merchandise **14** is replaced back onto tray **12**, inductive coil underneath tray **12** detects presence of the inductive coil of article of merchandise **14**. In response to detecting the inductive coil of article of merchandise **14**, an electrical signal is transmitted to the processor (which may be housed underneath tray **12**, within pedestal **20**, or a remote location). Upon receipt of this signal, the processor actuates electric motor **36** to raise locking block **28**, closing arm **18**.

(39) FIGS. **5**B-C and **6**B-C depict that electric motor **36** is screw-threadedly connected to locking block **28** via a driving screw **38**. In an embodiment, electric motor **36** may be a worm gear motor. When actuated, electric motor **36** causes driving screw **38** to rotate, which causes locking block **28** to rise. Locking block **28** may have a finger **39** configured to translate up and down within a corresponding groove disposed within pedestal **20**. In this manner, finger **39** restricts locking block **28** against rotation relative to pedestal **20** and, thus, locking block **28** can only move up or down. Thus, when the presence of article of merchandise **14** on tray **12** is detected, electric motor **36** rotates driving screw **38**, causing locking block **28** to move upward and engage stem **30** of arm **18**. The point of engagement between locking block **38** and stem **30** is offset relative to pivot axis **32**, thereby creating a moment on stem **30**, urging arm **18** to rotate into the closed position. The force exerted onto stem **30** by locking member **28** is greater than the force exerted onto stem **30** by biasing member **34**. Accordingly, as locking block **28** rises, stem **30** rotates about pivot axis **32**. In this manner, as locking block **28** rises, arm **18** is transitioned into the closed position.

(40) Another feature of anti-theft device **10** depicted in FIGS. **5**B-C and **6**B-C pertains to a blocking screw **40**, which is screw-threadedly coupled to locking block **28**. When locking block **28** is raised, blocking screw **40** aligns with access port **22** and blocks access to locking member **24**. When locking block **28** is lowered, blocking screw **40** is retracted away from access port **22**, enabling an authorized user to access locking member **24** by inserting a tool into access port **22**. In this manner, blocking screw **40** ensures that locking member **24** can be accessed only when locking block **28** is lowered and electromechanically actuated arm **18** is in the opened position, thus creating an additional layer of security.
(41) Another feature of anti-theft device **10** pertains to an ability to remove article of merchandise **14** by an authorized user in case electric motor **36** malfunctions or there is a power outage. If such event were to occur, pedestal **20** can be unmounted from the display surface, revealing access to the head of blocking screw **40**. At this point, the authorized user can use a tool—for example, a screwdriver—to manually retract blocking screw **40** away from access port **22**. After blocking screw **40** has been sufficiently retracted from access port **22**, the user can access and operate locking member **24** to retract its teeth away from the teeth of arms **16** and **18**, as depicted in FIGS. **3**A-B. With locking member **24** in the retracted position, the user can extend arms **16** and **18** relative to tray **12**, releasing article of merchandise **14**.
(42) In an embodiment, the processor may be configured to detect the rate of rotation of driving screw **38** (pulse counting). If an obstacle interferes with arm **18** as it is being transitioned into the closed position, the rate of rotation of driving screw **38**—and, hence, electric motor **36**—will decrease. Upon detecting the decreased rate of rotation, the processor may be configured to stop electric motor **36**, electric motor **36** can be reversed, and/or an alarm may be produced. This feature has a two-fold benefit: (1) safety: if a user's finger or clothing gets caught between tray **12** and arm **18**, shutting off electric motor **30** would prevent a potential injury; and (2) security, if someone interferes with normal operation of arm **18**, anti-theft device **10** will alert the store personnel.
(43) Upon completion of the customer interaction, article of merchandise **14** must be placed back into anti-theft device **10** by laterally sliding article of merchandise **14** into the grips of arms **16**. When article of merchandise **14** is properly placed onto tray **12**, the inductive coil of tray **12** will detect inductive coil of article of merchandise **14**, thereby automatically actuating electric motor **36** to raise locking block **28** and to transition electromechanically actuated arm **18** into the closed position.
(44) An advantage of using the electromechanical mechanism described above is that, because pivot axis **32** of stem **30** is not aligned with the axis of rotation of electric motor **36**, the worm-gear engagement between electric motor **36** and electromechanically actuated arm **18** ensures that arm **18** cannot be opened using manual force. Furthermore, preferably, electromechanically actuated arm **18** is made of a metal or a metal alloy and, therefore, is not susceptible to being broken or deformed using manual force or basic manual tools. Thus, because arm **18** cannot be forced into the open position and cannot be broken or bent, article of merchandise **14** cannot be removed from tray **12** without operating the electromechanical mechanism, ensuring high level of anti-theft security.
(45) The advantages set forth above, and those made apparent from the foregoing description, are efficiently attained. Since certain changes may be made in the above construction without departing from the scope of the invention, it is intended that all matters contained in the foregoing description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. An anti-theft device for securing an article of merchandise, comprising: a pedestal affixed to a support surface; an electromechanically operated arm configured to selectively immobilize the article of merchandise relative to the pedestal, wherein the electromechanically operated arm has a

closed position in which the article of merchandise is immobilized relative to the pedestal and an open position in which the article of merchandise can be removed from the pedestal; a locking block movingly disposed within the pedestal, wherein the locking block has a first position relative to the pedestal in which the locking block is configured to immobilize the electromechanically operated arm in the closed position, and wherein the locking block has a second position relative to the pedestal in which the locking block is configured to release the electromechanically operated arm, enabling the electromechanically operated arm to transition into the open position, an electrical actuator operatively connected to a locking block which is operatively engaged with the electromechanically operated arm, wherein the locking block is configured to move between the first position and the second position under an influence of a force produced by the electrical actuator thereby selectively closing or opening the electromechanically operated arm, wherein in the closed position the electromechanically operated arm is restricted from being transitioned into the open position via application of a manual force or moment thereto; and a processor in electrical communication with the electrical actuator, wherein in response to a designated signal, the processor is configured to energize the electrical actuator causing the electromechanically operated arm to move from the closed position into the open position thereby releasing the article of merchandise for removal from the pedestal.
2. The anti-theft device of claim 1, wherein the electromechanically operated arm is configured to rotate about a pivot axis.
3. The anti-theft device of claim 2, wherein the locking block is configured to translate in a plane substantially perpendicular to the pivot axis.
4. The anti-theft device of claim 1, further comprising a plurality of arms configured to extend and retract relative to the pedestal.
5. The anti-theft device of claim 4, wherein the plurality of arms is configured to restrict the article of merchandise to linear movement solely in a direction of the electromechanically operated arm.
6. The anti-theft device of claim 4, wherein at least some of the plurality of arms have teeth disposed thereon, and wherein a locking member is configured to selectively engage the teeth thereby immobilizing the plurality of arms relative to the pedestal or to selectively disengage the teeth thereby enabling the plurality of arms to be extended relative to the pedestal.
7. The anti-theft device of claim 6, wherein the locking member is operable via a port disposed on the pedestal, and wherein the port is occluded when the electromechanically operated arm is in the closed position thereby precluding operation of the locking member.
8. The anti-theft device of claim 1, wherein the anti-theft device is configured to generate an alarm responsive to detecting an obstacle obstructing the electromechanically operated arm during operation thereof.
9. The anti-theft device of claim 1, wherein the processor is configured to deactivate the electrical actuator responsive to detecting an obstacle obstructing the electromechanically operated arm during operation thereof.
10. The anti-theft device of claim 1, wherein the designated signal is transmitted wirelessly.
11. The anti-theft device of claim 1, wherein the designated signal is transmitted using a near field communication (NFC), a radio frequency (RF), or an optical signal.
12. The anti-theft device of claim 1, wherein the designated signal is transmitted responsive to receiving data associated with a credit card, an identification card, a driver's license, a token, a proprietary card, a chip, biometric information, or an input of user credentials.
13. A method of securing an article of merchandise against theft, comprising: mounting a pedestal to a support surface, wherein an electromechanically operated arm is pivotally disposed relative to the pedestal and has a closed position in which the electromechanically operated arm immobilizes the article of merchandise relative to the pedestal and an open position in which the electromechanically operated arm is pivoted away from the article of merchandise thereby enabling removal thereof from the pedestal; placing the article of merchandise onto the pedestal while the

electromechanically operated arm is in the open position; transitioning the electromechanically operated arm into the closed position, wherein the electromechanically operated arm is restricted by a locking block from being transitioned back into the open position via a manual force or moment applied thereto, wherein the locking block is movingly disposed within the pedestal and has a first position relative to the pedestal in which the locking block immobilizes the electromechanically operated arm in the closed position, and wherein the locking block has a second position relative to the pedestal in which the locking block releases the electromechanically operated arm, enabling the electromechanically operated arm to transition into the open position; and transmitting a designated signal responsive to which a processor is configured to energize an electrical actuator operatively connected to the locking block, wherein the electromechanically operated arm is configured to move from the closed position into the open position in response to the locking block being transitioned into the second position under an influence of a force produced by the electrical actuator, thereby releasing the article of merchandise for removal from the pedestal.
14. The method of claim 13, wherein the designated signal is transmitted wirelessly.
15. The method of claim 14, wherein the designated signal is transmitted using a near field communication (NFC), a radio frequency (RF), or an optical signal.
16. The method of claim 13, wherein the electromechanically operated arm is configured to rotate about a pivot axis, and wherein the locking block is configured to translate in a plane substantially perpendicular to the pivot axis.
17. The anti-theft device of claim 1, further comprising an electronic circuitry disposed within the pedestal configured to detect that the article of merchandise has been placed onto the pedestal.
18. The anti-theft device of claim 17, wherein the electronic circuitry comprises a first inductive coil configured to detect a second inductive coil disposed within the article of merchandise.
19. The method of claim 13, wherein the pedestal comprises an electronic circuitry configured to detect that the article of merchandise has been placed onto the pedestal.
20. The anti-theft device of claim 19, wherein the electronic circuitry comprises a first inductive coil configured to detect a second inductive coil disposed within the article of merchandise.
